UNIVERSITY NAME

Analysis V — Final Exam

Year: 2022

Exercise 1:

Let $E = C^1([0,1], \mathbb{R})$, the space of C^1 functions from $[0,1]$ to $\mathbb{R}$. For $f \in E$, define

$$N(f) = \|f\|_\infty + \|f'\|_\infty \quad \text{with} \quad \|f\|_\infty = \sup_{x\in[0,1]} |f(x)|.$$

1. Show that N is a norm on E. Is it equivalent to $\|\cdot\|_\infty$?
2. (a) Verify that $N'(f) = |f(0)| + \|f'\|_\infty$ is a norm on E.
   (b) Show that it is equivalent to N.

Answer Area

1. **Norm properties:**
   - **Positivity**: $N(f) \geq 0$. If $N(f) = 0$, then $\|f\|_\infty = 0$ and $\|f'\|_\infty = 0$, so $f = 0$.
   - **Homogeneity**: $N(\lambda f) = |\lambda|(\|f\|_\infty + \|f'\|_\infty) = |\lambda| N(f)$.
   - **Triangle inequality**: $N(f+g) \leq \|f\|_\infty + \|g\|_\infty + \|f'\|_\infty + \|g'\|_\infty = N(f) + N(g)$.

   Thus N is a norm.

   Equivalence with $\|\cdot\|_\infty$: They are not equivalent. Consider $f_n(x) = \sin(n\pi x)$. Then:

   $$\|f_n\|_\infty = 1, \quad \|f_n'\|_\infty = n\pi, \quad \text{so } N(f_n) \sim n\pi \to \infty$$

   while $\|f_n\|_\infty$ remains bounded. No constant C exists such that $N(f) \leq C\|f\|_\infty$ for all f.

2. (a) **Norm properties for** N':
   - **Positivity**: Clear. If $N'(f) = 0$, then $|f(0)| = 0$ and $\|f'\|_\infty = 0$ so $f' = 0 \implies f$ constant, and with $f(0) = 0 \implies f = 0$.
   - **Homogeneity**: $N'(\lambda f) = |\lambda f(0)| + \|\lambda f'\|_\infty = |\lambda| N'(f)$.
   - **Triangle inequality**: $N'(f+g) \leq |f(0) + g(0)| + \|f' + g'\|_\infty \leq N'(f) + N'(g)$.

   (b) **Equivalence with** N: Show $\exists c, C > 0$ such that $cN(f) \leq N'(f) \leq CN(f)$.

- **Upper bound**: $N'(f) = |f(0)| + \|f'\|_\infty \leq \|f\|_\infty + \|f'\|_\infty = N(f)$.
- **Lower bound**: For any $x \in [0,1]$, $f(x) = f(0) + \int_0^x f'(t)dt$, so

$$|f(x)| \leq |f(0)| + \int_0^1 |f'(t)|dt \leq |f(0)| + \|f'\|_\infty.$$

Taking supremum: $\|f\|_\infty \leq |f(0)| + \|f'\|_\infty = N'(f)$. Thus $N'(f) \leq N(f) \leq 2N'(f)$ (since $N(f) = \|f\|_\infty + \|f'\|_\infty \leq N'(f) + \|f'\|_\infty \leq 2N'(f)$).

So N and N' are equivalent with constants $c = 1/2$, $C = 1$.

Exercise 2:

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous function and I an open interval in $\mathbb{R}$. Show that $E = \{x \in \mathbb{R}^n : f(x) \in I\}$ is an open subset of $\mathbb{R}^n$.

Answer Area

Let $x_0 \in E$. Then $f(x_0) \in I$. Since I is open, there exists $\varepsilon > 0$ such that

$$(f(x_0) - \varepsilon, f(x_0) + \varepsilon) \subset I.$$

Since f is continuous at x_0, there exists $\delta > 0$ such that for all $x \in \mathbb{R}^n$ with $\|x - x_0\| < \delta$, we have

$$|f(x) - f(x_0)| < \varepsilon.$$

This implies $f(x) \in (f(x_0) - \varepsilon, f(x_0) + \varepsilon) \subset I$. Therefore, for all x in the open ball $B(x_0, \delta)$, we have $f(x) \in I$, i.e., $x \in E$.
Thus, every point $x_0 \in E$ has an open neighborhood contained in E, so E is open in $\mathbb{R}^n$.

Exercise 3:

1. Calculate

$$\lim_{(x,y)\to(0,0)} \frac{\sin(xy)}{x^2 + y^2}$$

2. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{x^2y^2}{\sqrt{(x^2+y^2)^3}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

   (a) Show that $\nabla f(0,0) = (0,0)$.

   (b) Show that the function $\frac{\partial f}{\partial x} : \mathbb{R}^2 \to \mathbb{R}$ is not continuous at $(0,0)$.

3. Let U be an open subset of $\mathbb{R}^2$, $(u, v) \in U$ with $u \neq 0$, and $f : U \to \mathbb{R}$ a C^1 function, homogeneous of degree α, such that:

$$f(u, v) = 0 \quad \text{and} \quad \frac{\partial f}{\partial y}(u, v) \neq 0$$

Give explicitly the implicit function $y = \psi(x)$ that f defines near the point u and which satisfies $\psi(u) = v$.

Answer Area

1. Use polar coordinates: $x = r\cos\theta, y = r\sin\theta$. Then

$$\frac{\sin(xy)}{x^2 + y^2} = \frac{\sin(r^2\cos\theta\sin\theta)}{r^2}.$$

For small t, $\sin t \sim t$, so

$$\frac{\sin(r^2\cos\theta\sin\theta)}{r^2} \sim \frac{r^2\cos\theta\sin\theta}{r^2} = \cos\theta\sin\theta.$$

The limit as $r \to 0$ depends on θ: it equals $\cos\theta\sin\theta$, which varies with θ. For example:

- Along $x = 0$ or $y = 0$: limit $= 0$.
- Along $y = x$: $\cos\theta\sin\theta = \frac{1}{2}$ when $\theta = \pi/4$.

Since the limit depends on the direction, it does not exist.

2. (a) Compute partial derivatives at $(0, 0)$:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h\to 0}\frac{f(h, 0) - f(0, 0)}{h} = \lim_{h\to 0}\frac{0}{h} = 0.$$

Similarly $\frac{\partial f}{\partial y}(0, 0) = 0$. Thus $\nabla f(0, 0) = (0, 0)$.

(b) For $(x, y) \neq (0, 0)$, compute $\frac{\partial f}{\partial x}$ using quotient rule. Let $r = \sqrt{x^2 + y^2}$, then

$$f(x, y) = \frac{x^2y^2}{r^3}.$$

Compute derivative (or check along a path). Consider approaching $(0, 0)$ along $y = x$: For $y = x$, $f(x, x) = \frac{x^4}{(2x^2)^{3/2}} = \frac{x^4}{2^{3/2}|x|^3} \sim \frac{|x|}{2^{3/2}}$. The partial derivative $\frac{\partial f}{\partial x}$ along this path does not tend to 0 (shows discontinuity). More formally: compute $\frac{\partial f}{\partial x}(x, y)$ explicitly and show its limit as $(x, y) \to (0, 0)$ does not exist or does not equal 0.

3. Since f is homogeneous of degree α, Euler's theorem gives:

$$x\frac{\partial f}{\partial x}(x, y) + y\frac{\partial f}{\partial y}(x, y) = \alpha f(x, y).$$

At (u, v), $f(u, v) = 0$, so

$$u\frac{\partial f}{\partial x}(u, v) + v\frac{\partial f}{\partial y}(u, v) = 0.$$

By the Implicit Function Theorem, since $\frac{\partial f}{\partial y}(u,v) \neq 0$, there exists ψ near u with $\psi(u) = v$ and $f(x, \psi(x)) = 0$.

Differentiate $f(x, \psi(x)) = 0$:

$$\frac{\partial f}{\partial x}(x, \psi(x)) + \frac{\partial f}{\partial y}(x, \psi(x))\psi'(x) = 0.$$

At $x = u$, $\psi(u) = v$:

$$\psi'(u) = -\frac{\frac{\partial f}{\partial x}(u,v)}{\frac{\partial f}{\partial y}(u,v)}.$$

Using Euler's relation: $\frac{\partial f}{\partial x}(u,v) = -\frac{v}{u}\frac{\partial f}{\partial y}(u,v)$ (since $u\frac{\partial f}{\partial x} + v\frac{\partial f}{\partial y} = 0$), so

$$\psi'(u) = \frac{v}{u}.$$

The linear approximation near u is:

$$\psi(x) \approx v + \frac{v}{u}(x - u).$$

For the exact implicit function, if f is homogeneous, one can show $\psi(x) = \frac{v}{u}x$ (if f is linear in y, but generally it's the tangent approximation).